



VEHICLE SERVICE CENTRE MANAGEMENT SYSTEM



MINI PROJECT REPORT

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In partial fulfilment of the requirement for

the award of the degree

of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING

THE KAVERY ENGINEERING COLLEGE

(An Autonomous Institution, affiliated to Anna University Chennai and Approved by AICTE,
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MECHERI, SALEM – 636453

MAY 2026

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BONAFIDE CERTIFICATE

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ACKNOWLEDGEMENT

We wish to express our sincere gratitude to our Advisor **Dr. A. K. NATESAN**, our Chairman Thiru **A. ANBALAGAN**, our Secretary Prof **S. K. ELANGO VAN** for providing immense facilities at our institution.

We would like to acknowledge our Principal **Dr. V. DURAISAMY M.E., Ph.D., FIE**, for fostering a supportive learning environment that encourages collaboration and teamwork. Your leadership has set a positive example for all of us and has created a culture of academic excellence.

We wish to extend our heartfelt gratitude to our **Head of the Department Dr. M. BALAMURUGAN M.E., Ph.D.**, for your support and guidance throughout the completion of the thesis.

We are in deep gratitude to our department faculty who always been supporting us through thick and thin respected **Project Coordinator Dr. M. BALAMURUGAN M.E., Ph.D.**, Associate Professor for the engulf support for the project.

We are highly indebted to provide our heart full thanks to our **Project Guide Dr. M. BALAMURUGAN M.E., Ph.D.**, Associate Professor for his valuable ideas, encouragement and supportive guidance throughout the project.

We wish to extend our sincere thanks to all faculty members of our **Computer Science and Engineering** for guiding us and providing valuable feedback along the way for successful completion of this project.

We would like to express my sincere gratitude to all the students in my group for their hard work, dedication, and cooperation throughout the preparation and editing stages of the manuscript.

We would like to thank all those who have contributed to the success of our college project. Your support and guidance have been truly appreciated, and we could not have completed this project without your help.

ABSTRACT

A Vehicle Service Centre Management System is a web-based or software application designed to automate and manage the day-to-day operations of a vehicle service Centre. The system aims to replace manual processes with a digital platform that improves efficiency, accuracy, and overall service quality. It enables service Centre staff to handle customer registrations, vehicle details, service requests, job card creation, and billing in an organized manner. By maintaining a centralized database, the system ensures that all information related to customers, vehicles, service history, and spare parts is stored securely and can be accessed easily whenever required. This reduces paperwork, minimizes errors, and saves time for both employees and customers. Additionally, the system can generate reports, track service progress, and provide timely updates, enhancing customer satisfaction. Furthermore, the system includes modules for inventory management, employee management, and service scheduling, which help in optimizing resources and improving productivity. It allows administrators to monitor stock levels of spare parts, assign tasks to technicians, and manage service timelines efficiently. The integration of billing and payment features ensures accurate invoicing and transparent transactions.

The system can also support notifications and reminders for service appointments, improving customer engagement. Overall, the Vehicle Service Centre Management System provides a comprehensive solution that streamlines operations, reduces operational costs, and enhances the performance of the service Centre, making it suitable for modern automotive service industries. A Vehicle Service Centre Management System is an advanced software solution developed to simplify and automate the operations of automobile service Centres. It integrates various functionalities such as customer registration, vehicle tracking, service scheduling, job allocation, and billing into a single platform. The system reduces the dependency on manual record-keeping, thereby minimizing human errors and improving data accuracy. It enables service advisors to efficiently handle customer requests, assign tasks to technicians, and monitor service progress. By maintaining a well-structured database of customers and vehicles, the system ensures quick retrieval of information, which enhances service quality and builds customer trust. The digital approach also helps in reducing paperwork, saving time, and improving overall workflow efficiency within the organization.

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
	ABSTRACT	v
	TABLE OF CONTENT	vi
	LIST OF FIGURES	viii
	LIST OF TABLES	ix
	LIST OF ABBREVIATIONS	x
1	INTRODUCTION	1
2	OBJECTIVES	2
3	PROBLEM STATEMENT	3
4	MOTIVATION	4
5	PROJECT SPECIFICATIONS	5
	5.1 Hardware Requirements	5
	5.2 Software Requirements	6
	5.3 Technology Stack	6
6	FLOW DIAGRAM	7
7	UML DIAGRAMS	9
	7.1 Use Case Diagram	10
	7.2 Class Diagram	11
	7.3 Sequence Diagram	12
	7.4 Collaboration Diagram	13
	7.5 Package Diagram	14
8	MODULES	15
	8.1 Login Module	16
	8.2 Dashboard Module	17
	8.3 Customer Management Module	18

	8.4 Vehicle Management Module	19
	8.5 Service Management Module	20
	8.6 Billing Module	21
9	TESTING	22
	9.1 Test Case Generation	23
	9.2 Black Box Testing	24
	9.3 White Box Testing	25
	9.4 Unit Testing	26
	9.5 Integration Testing	27
	9.6 System Testing	28
	9.7 Acceptance Testing	29
10	CONCLUSION	30
11	REFERENCES	31

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
6	Flow Diagram of Vehicle Service Centre Management System	8
7.1	Use Case Diagram of Vehicle Service Centre Management System	10
7.2	Class Diagram of Vehicle Service Centre Management System	11
7.3	Sequence Diagram of Vehicle Service Centre Management System	12
7.4	Collaboration Diagram of Vehicle Service Centre Management System	13
7.5	Package Diagram of Vehicle Service Centre Management System	14
8	Vehicle Service Centre Management System	15
8.1	Login Module	16
8.2	Dashboard Module	17
8.3	Customer Management Module	18
8.4	Vehicle Management Module	19
8.5	Service Management Module	20
8.6	Billing Module	21

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
5.1	Hard Requirements	5
5.2	Software Requirements	6
5.3	Technology Stack	6
9.1	Sample Test Cases	23
9.2	Sample Black Box Test Cases	24
9.3	Sample White Box Test Cases	25
9.4	Sample Unit Test Cases	26
9.5	Sample Integration Test Cases	27
9.6	Sample System Test Cases	28
9.7	Sample Acceptance Test cases	29

LIST OF ABBREVIATIONS

SQL	Structured Query Language
UML	Unified Modelling Language
UI	User Interface
API	Application Programming Interface
RAM	Random Access Memory
URL	Uniform Resource Locator
DB	Data Base
JS	Java Script
CPU	Central Processing Unit
PDF	Portable Document Format
ID	Identification
HTML	Hyper Text Markup Language
CSS	Cascading Style Sheets
VS Code	Visual Studio Code
HTTPS	Hyper Text Transfer Protocol Secure
HTTP	Hyper Text Transfer Protocol
OS	Operating System

Chapter 1

INTRODUCTION

The Vehicle Service Centre Management System is designed to provide a user-friendly interface for managing all service Centre activities efficiently. To begin with, the system must be properly installed and configured on a computer or server with necessary software requirements such as a web browser and database support. The administrator is responsible for creating user accounts for staff members, including service advisors, technicians, and billing personnel. Each user is assigned specific roles and permissions to ensure secure access to the system. Once logged in, users can navigate through different modules such as customer management, service requests, inventory, and billing. Proper training should be provided to staff to ensure they understand how to use each feature effectively and maintain accurate data entry. For daily operations, service advisors should register customer details and vehicle information when a new service request is received. They must create a job card that includes the type of service required, estimated cost, and expected delivery time. This job is then assigned to a technician, who updates the status of the service as work progresses.

The system allows real-time tracking of vehicle servicing, ensuring transparency and better coordination among staff. The inventory module should be regularly updated to reflect the availability of spare parts, and any usage during servicing must be recorded immediately. Billing staff can generate invoices automatically based on the services provided and parts used, ensuring accurate and efficient payment processing. To maintain system efficiency and reliability, regular updates and data backups are essential. The administrator should monitor system performance and ensure that all modules are functioning correctly. Security measures such as password protection and data encryption should be implemented to safeguard sensitive information. Periodic reports generated by the system should be reviewed to analyze business performance, customer satisfaction, and employee productivity. In case of any technical issues, proper troubleshooting steps or technical support should be followed. By adhering to these instructions, the Vehicle Service Centre Management System can be effectively utilized to enhance operational efficiency, reduce errors, and deliver high-quality service to customers.

Chapter 2

OBJECTIVES

The primary objective of the Vehicle Service Centre Management System is to automate and simplify the overall operations of a vehicle service Centre. The system aims to replace manual processes with a computerized solution that improves accuracy, reduces paperwork, and saves time. It focuses on maintaining detailed records of customers, vehicles, and service history in a centralized database for easy access and management. By providing a structured platform, the system helps service advisors efficiently handle customer requests, schedule services, and assign tasks to technicians. This leads to better coordination among staff members and ensures smooth workflow within the service Centre. Another important objective of the system is to enhance customer satisfaction by providing faster and more reliable services. The system enables online appointment booking, reducing waiting time and improving convenience for customers. It also ensures transparency in billing by generating accurate invoices based on services and spare parts used. Customers can receive timely updates about the status of their vehicles, which builds trust and improves communication.

Additionally, the system helps in maintaining service quality by tracking previous service records and ensuring that all customer requirements are properly addressed. The system also aims to improve business management and decision-making for service Centre owners. It provides features such as inventory management to track spare parts and avoid shortages or overstocking. The reporting module generates detailed reports on daily operations, revenue, employee performance, and service trends. These insights help management analyze performance and plan future strategies effectively. Furthermore, the system ensures data security and reliability through controlled access and regular backups. Overall, the objective of the Vehicle Service Centre Management System is to increase efficiency, improve service quality, and support the growth of the business through modern technological solutions.

Chapter 3

PROBLEM STATEMENT

The existing vehicle service Centre operations largely depend on manual processes, which create several inefficiencies and challenges. Service records are often maintained in paper files or basic spreadsheets, making it difficult to store, retrieve, and manage data effectively. This leads to issues such as misplaced records, data duplication, and increased chances of human error. Additionally, manual appointment handling causes scheduling conflicts and long waiting times for customers. Communication gaps between service advisors, technicians, and customers further reduce service quality and transparency. As the number of customers increases, managing these operations manually becomes more complex and time-consuming, ultimately affecting the overall performance of the service Centre.

Another major problem is the lack of proper inventory and billing management in traditional systems. Service Centres often struggle to keep track of spare parts availability, leading to either shortages or overstocking. Without an efficient tracking system, it becomes difficult to monitor the usage of parts and maintain accurate stock levels. Billing processes performed manually are prone to calculation errors, lack of transparency, and delays in generating invoices. This not only affects the financial accuracy of the organization but also creates dissatisfaction among customers. Moreover, the absence of centralized data makes it difficult for management to analyze business performance, track employee productivity, and make informed decisions.

There is also a significant need for improved customer service and digital accessibility in modern service Centres. Customers expect convenient features such as online appointment booking, real-time service updates, and quick access to their vehicle service history. However, traditional systems fail to provide these facilities, resulting in poor customer experience and reduced trust. The lack of automation and integration among different departments leads to delays, miscommunication, and inefficiency. Therefore, there is a clear need for a comprehensive Vehicle Service Centre Management System that can address these problems by automating operations, improving data management, enhancing communication, and delivering better service quality in a fast and reliable manner.

Chapter 4

MOTIVATION

The motivation behind developing the Vehicle Service Centre Management System arises from the increasing demand for efficient, organized, and user-friendly solutions in managing vehicle servicing operations. In many service Centres, traditional methods such as manual record-keeping and paper-based systems are still widely used. These methods often lead to data loss, confusion, and delays in handling customer requests. As the number of vehicles and customers continues to grow, it becomes difficult to manage service records, bookings, and vehicle details effectively without a proper digital system.

Another important factor that motivated this project is the need for systematic management of different activities within a service centre. A vehicle service centre typically handles multiple operations such as customer registration, vehicle tracking, service scheduling, and billing. Without a centralized system, managing these tasks becomes time-consuming and prone to errors. This project aims to integrate all these functionalities into a single platform, ensuring smooth and efficient workflow management.

The requirement for role-based access also plays a significant role in the development of this system. Different users such as administrators, staff, and customers have different responsibilities and access needs. Without proper role management, there is a risk of unauthorized access and misuse of data. By implementing a structured login system, the project ensures that each user can access only the relevant features, thereby improving security and operational efficiency.

Data security and reliability are also key motivating factors. Service centres handle important information such as customer details, vehicle information, and service history. If this data is not properly stored and protected, it may lead to serious issues such as data loss or unauthorized access. The system incorporates secure login mechanisms and proper data handling techniques to ensure that all information is safe and reliable.

Another motivation for this project is to reduce manual errors and improve the accuracy of service operations. Manual entry of service records and billing details often leads to mistakes, which can affect customer satisfaction and business performance. By automating these processes, the system ensures accurate record maintenance, faster service processing, and better overall efficiency.

Chapter 5

PROJECT SPECIFICATIONS

The project specification defines the hardware and software requirements needed for the development and execution of the Vehicle Service Centre Management System. It provides a clear overview of the system environment, tools, technologies, and resources used in implementing the project. These specifications ensure that the application can be developed, tested, deployed, and maintained efficiently in a proper technical environment. The hardware requirements describe the minimum system configuration required to run the application smoothly, while the software requirements and technology stack explain the tools, frameworks, database, and platforms used in building the system.

5.1 Hardware Requirements

The hardware requirements define the minimum physical resources needed to develop and run the Vehicle Service Centre Management System. A computer with a suitable processor, sufficient RAM, and adequate storage is required for smooth development, testing, and execution. A 64-bit operating system is recommended for better performance and compatibility. Internet connectivity is necessary for downloading tools, updates, and for future deployment purposes.

S.No	Requirements
1	Processor: Intel Core i3 or above
2	RAM: 4 GB
3	Storage: 128 GB or above
4	System Type: 64-bit Operating System
5	Internet Connection: Required for deployment and access

Table 5.1: Hardware Requirements

5.2 Software Requirements

The software requirements describe the tools and technologies used to develop and run the Vehicle Service Centre Management System. This project uses web-based technologies for frontend design and scripting for functionality. These tools help in creating an interactive and user-friendly application.

S.No	Requirements
1	Operating System: Windows 10/11
2	Programming Language: JavaScript
3	Scripting: JavaScript
4	Frontend: HTML, CSS
5	Editor: VS code/Notepad
6	Browser: Google Chrome
7	Version Control: Git (optional)

Table 5.2: Software Requirements

5.3 Technology Stack

The technology stack represents the core technologies used in the project. It includes the frontend for user interface design and scripting for implementing system functionality. Since this is a mini project, it uses a simple and lightweight architecture without a backend database.

Layer	Technology Used
Frontend	HTML, CSS
Scripting	JavaScript
Data Storage	Browser (Local/Temporary Storage)
Authentication	Basic Logic Validation (JavaScript)
Deployment	Local Browser

Table 5.3: Technology Stack

Chapter 6

FLOW DIAGRAM

The flow diagram of the Vehicle Service Centre Management System represents the step-by-step process that takes place within the system, starting from user access to the completion of service-related operations. It provides a clear understanding of how data flows through different modules and how the system manages user interactions. The diagram visually explains the sequence of operations and decision-making processes involved in authentication, navigation, and execution of various functionalities.

The system begins when a user accesses the application through a web browser. Once the website is opened, the system redirects the user to the login page. This ensures that only authorized users can access the system and prevents unauthorized usage. The login page acts as the entry point for all users.

At the login stage, the user is required to enter a username and password. The system verifies these credentials using basic authentication logic. If the entered details are incorrect, the system displays an error message and asks the user to re-enter the correct information. This validation step ensures that only authenticated users can proceed further into the system.

Once the login is successful, the user is directed to the main dashboard. This dashboard acts as the central control panel where users can navigate to different modules such as Customer Management, Vehicle Management, and Service Booking. This step is important because it provides organized access to all system functionalities.

In the Customer Management module, the flow allows the user to add new customer details such as name and contact information. The system stores this data and displays it in a list format. Users can view the stored customer records, which helps in managing customer information efficiently.

In the Vehicle Management module, the user can enter vehicle details such as vehicle model and registration number. The system saves this information and displays it for future reference. This ensures that each vehicle is properly tracked within the system.

In the Service Booking module, the flow continues with booking vehicle services. The user selects a service type and date, and the system records the booking details. Once the booking is confirmed, it is displayed in a list, allowing users to keep track of all service requests.

Another important part of the flow diagram is data handling. Whenever a new customer, vehicle, or booking is added, the system immediately updates and displays the information. This ensures that all records are maintained accurately and are easily accessible. The system also ensures smooth navigation between different modules. Users can move from one section to another using the dashboard menu, which improves usability and efficiency.

Finally, after completing all tasks, the user logs out of the system. The system then ends the session and redirects the user back to the login page. This step ensures that the system remains secure and prevents unauthorized access after logout. Overall, the flow diagram clearly illustrates how the system manages user authentication and controls the flow of operations from login to logout. It ensures a smooth and structured workflow for managing customers, vehicles, and service bookings while maintaining simplicity, efficiency, and ease of use.

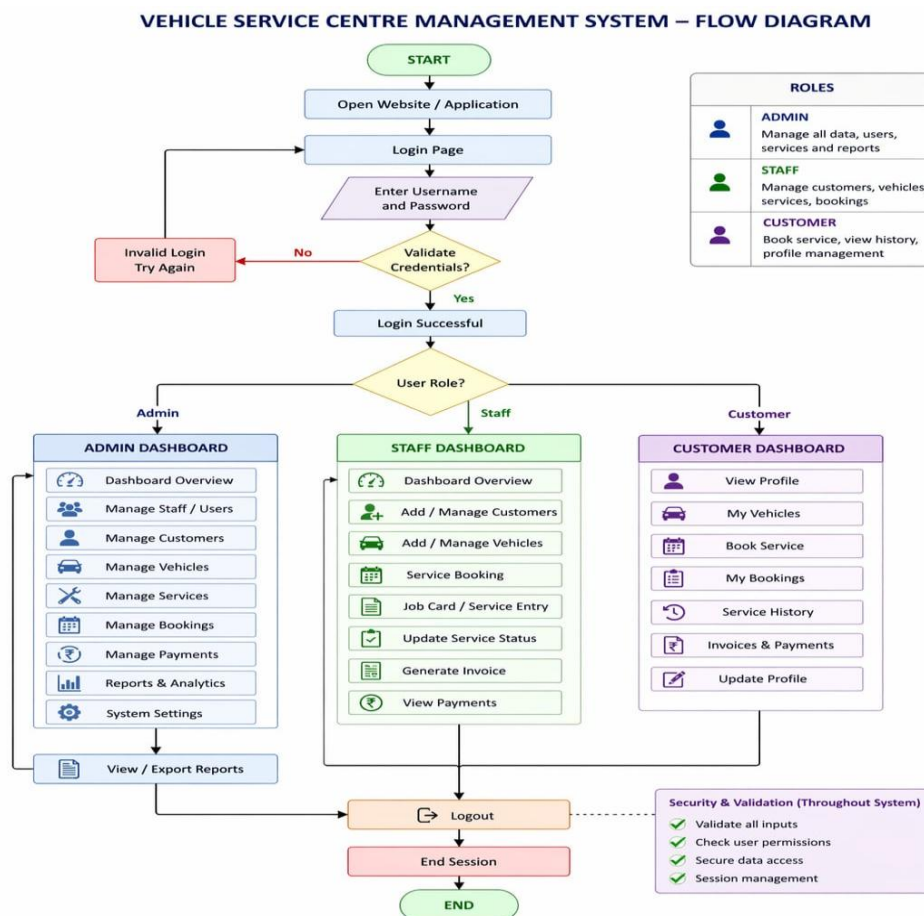


Figure 6: Flow Diagram of Vehicle Service Centre Management System

Chapter 7

UML DIAGRAMS

The UML diagram of the Vehicle Service Centre Management System provides a clear and structured representation of the system's design, architecture, and overall functionality. It plays a crucial role in visualizing how different components of the system interact with each other and how the entire system is organized using standard object-oriented modelling techniques. UML (Unified Modelling Language) diagrams help developers, designers, and stakeholders understand the system from both a functional and structural perspective. The diagram mainly includes the Use Case Diagram and the Class Diagram, which together explain both the external interactions of users with the system and the internal structure of the system components. The UML diagram of the Vehicle Service Centre Management System provides a clear and structured representation of the system's design and functionality. It helps in visualizing how different components interact and how the system is organized using standard modelling techniques. The diagram mainly includes the Use Case Diagram and the Class Diagram, which together explain both the external interactions and internal structure of the system.

The following UML diagrams are included in this report:

1. Use Case Diagram
2. Class Diagram
3. Sequence Diagram
4. Collaboration Diagram
5. Package Diagram

7.1 Use Case Diagram

The use case diagram represents the functional requirements of the Vehicle Service Centre Management System. It shows how different users interact with the system to perform various operations. The main actors in the system are Admin, Staff, and Customer. Each actor has specific roles and responsibilities within the system. The Customer can log in, manage vehicle details, book services, and view service history. The Staff handles customer records, vehicle registration, and service bookings. They also update service status and assist customers during service operations. The admin manages users, monitors activities, and controls overall system operations. The diagram clearly shows the relationship between users and system functions. It helps in understanding system behaviour and ensures proper representation of all functionalities.

The use case diagram of the Banking Management system is shown below.

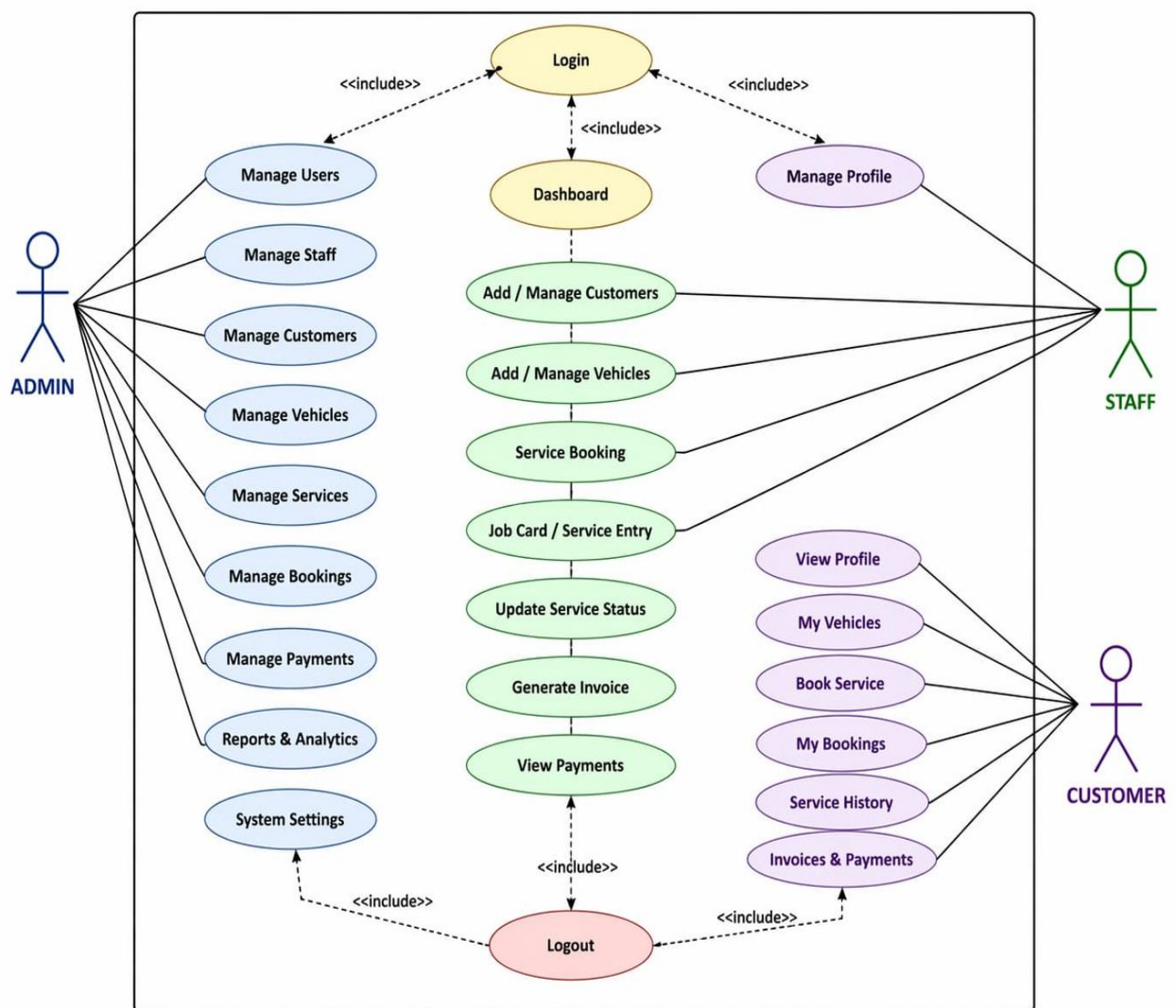


Figure 7.1: Use Case Diagram of Vehicle Service Centre Management System

7.2 Class Diagram

The class diagram represents the static structure of the Vehicle Service Centre Management System. It illustrates the key classes such as User, Customer, Vehicle, Service, Booking, and Payment, along with their attributes, methods, and relationships. Each class defines specific properties like user ID, name, contact details, vehicle number, service type, booking date, and payment status, as well as functions to manage these details. The diagram clearly shows how different entities are connected, for example, a customer can have multiple Vehicles, each Vehicle can have multiple Service records, and each Booking is linked to a Customer and a Vehicle. It also represents the interactions between classes through associations, ensuring proper data flow within the system. This diagram helps in understanding how the system is designed using object-oriented principles and how data is structured and maintained. Overall, it provides a clear blueprint for system development and ensures that all components are properly connected and organized.

The class diagram of the Banking Management System is shown below.

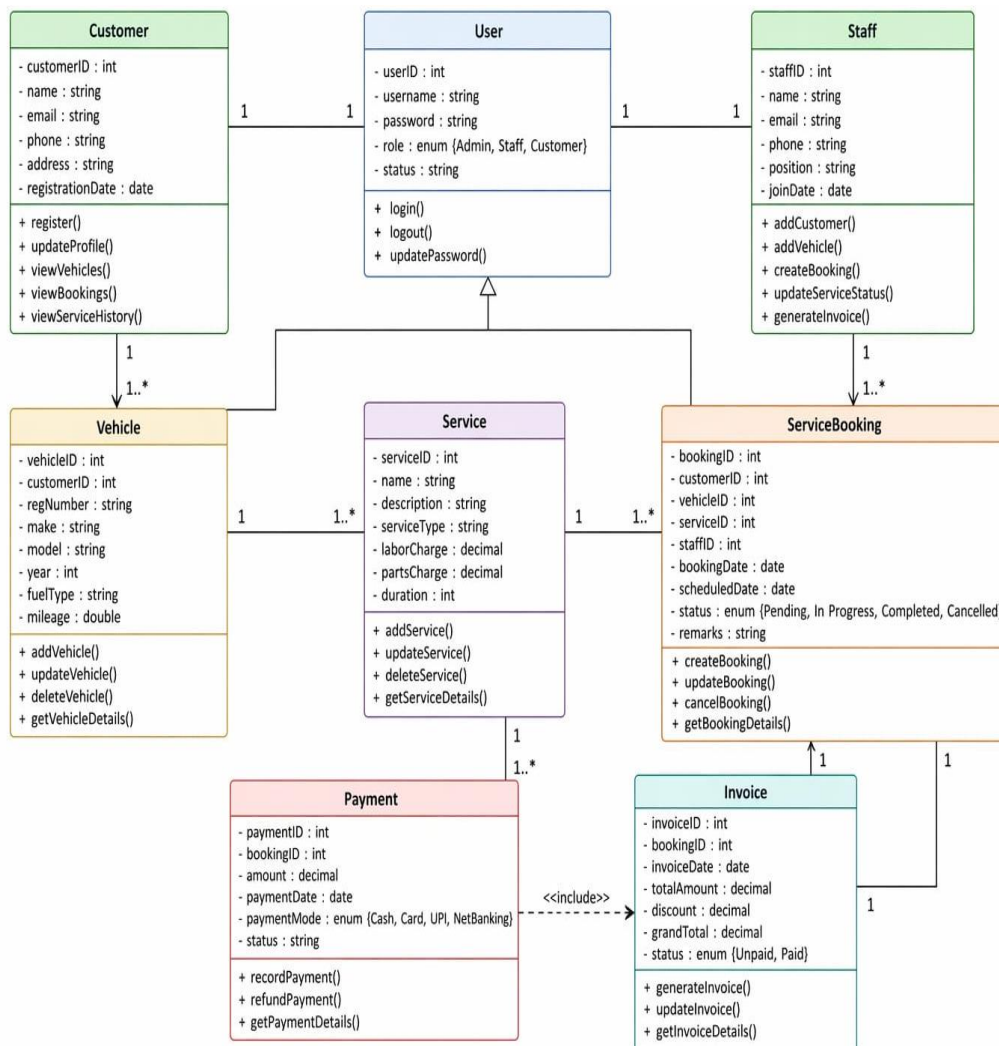


Figure 7.2: Class Diagram of Vehicle Service Centre Management System

7.3 Sequence Diagram

The sequence diagram represents the flow of interaction between different components of the system in a time-based sequence. It shows how messages are passed between users and system modules to complete operations such as login, transaction processing, and account management. In this project, it helps in understanding the step-by-step execution of banking activities.

The sequence diagram of the Banking Management System is shown below.

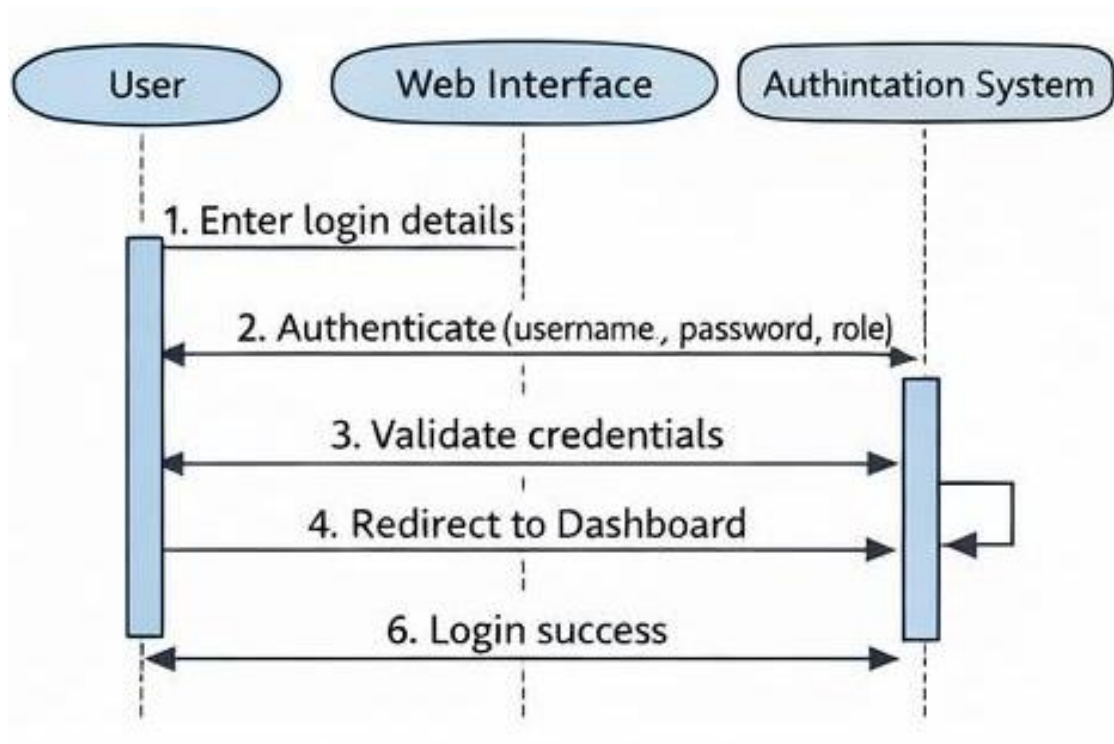


Figure 7.3: Sequence Diagram of Vehicle Service Centre Management System

7.4 Collaboration Diagram

The Collaboration Diagram (also known as a Communication Diagram in Unified Modelling Language) represents how different objects in the system communicate with each other to perform specific tasks. It focuses on the relationships and interactions between system components such as user, service, vehicle, and booking modules. This diagram helps in understanding how different parts of the system work together efficiently.

The collaboration diagram of the Vehicle Service Centre Management System is shown below.

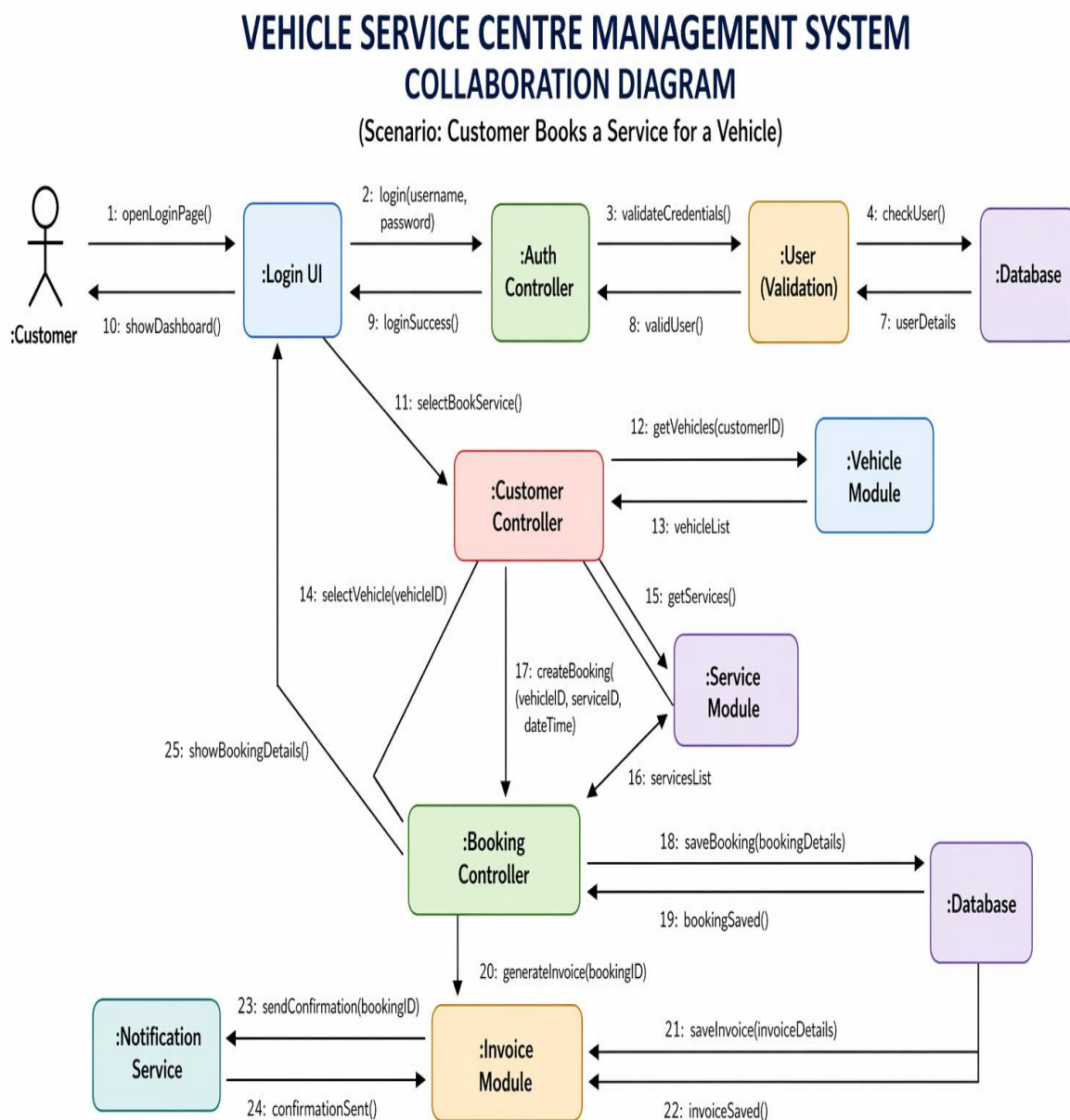


Figure 7.4: Collaboration Diagram of Vehicle Service Centre Management System

7.5 Package Diagram

The Package Diagram in Unified Modelling Language represents the overall organization of the Vehicle Service Centre Management System into different modules or packages. It shows how related components such as authentication, vehicle management, booking, service handling, and admin control are grouped together. This diagram helps in understanding the modular structure and layered architecture of the system.

The package diagram of the Vehicle Service Centre Management System is shown below.

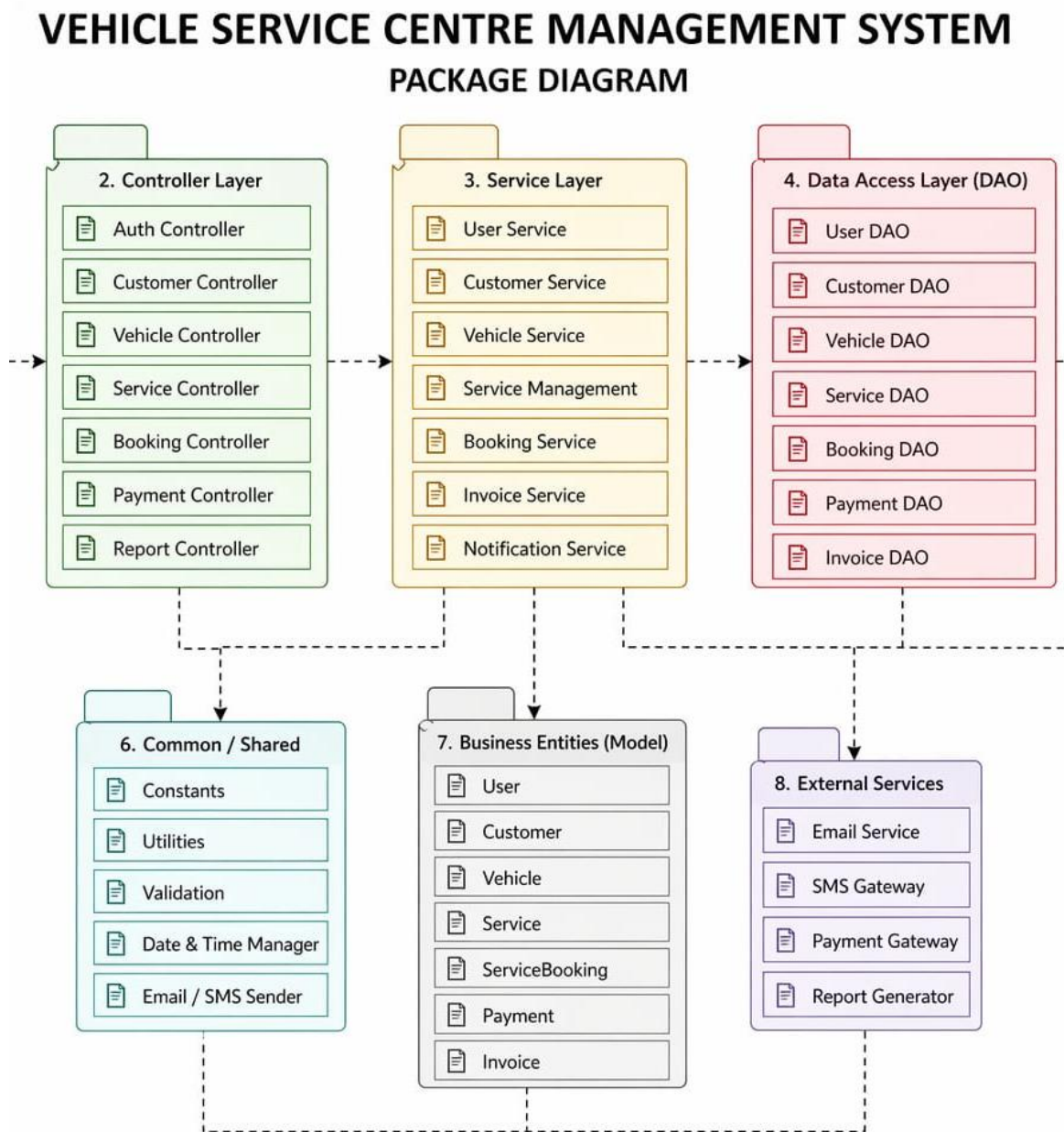


Figure 7.5: Package Diagram of Vehicle Service Centre Management System

Chapter 8

MODULES

The Vehicle Service Centre Management System is divided into multiple functional modules to ensure efficient handling of service operations and proper management of customer activities. Each module is responsible for a specific task and contributes to the overall functionality of the system. The modular approach helps in improving system organization, reducing complexity, and making the application easier to maintain and upgrade. These modules work together to provide a reliable, efficient, and user-friendly environment for customers and administrators.

The system consists of the following modules:

1. Login Module
2. Dashboard Module
3. Customer Management Module
4. Vehicle Management Module
5. Service Management Module
6. Billing Module



Figure 8 : Vehicle Service Centre Management System

8.1 Login Module

The Login Module is responsible for authenticating users before granting access to the system. Users must enter their username and password to log in. The system verifies the credentials and ensures that only authorized users can access the application. If the login is successful, the user is redirected to the dashboard page; otherwise, an error message is displayed. This module ensures security and controlled access to the system.

Sample code:

Script.js

```
function login() {  
  
    let username = document.getElementById("username").value;  
  
    let password = document.getElementById("password").value;  
  
    if (username === "admin" && password === "1234") {  
  
        window.location.href = "dashboard.html";  
  
    } else {  
  
        document.getElementById("error").innerText = "Invalid Login"}}}
```

Sample Output:

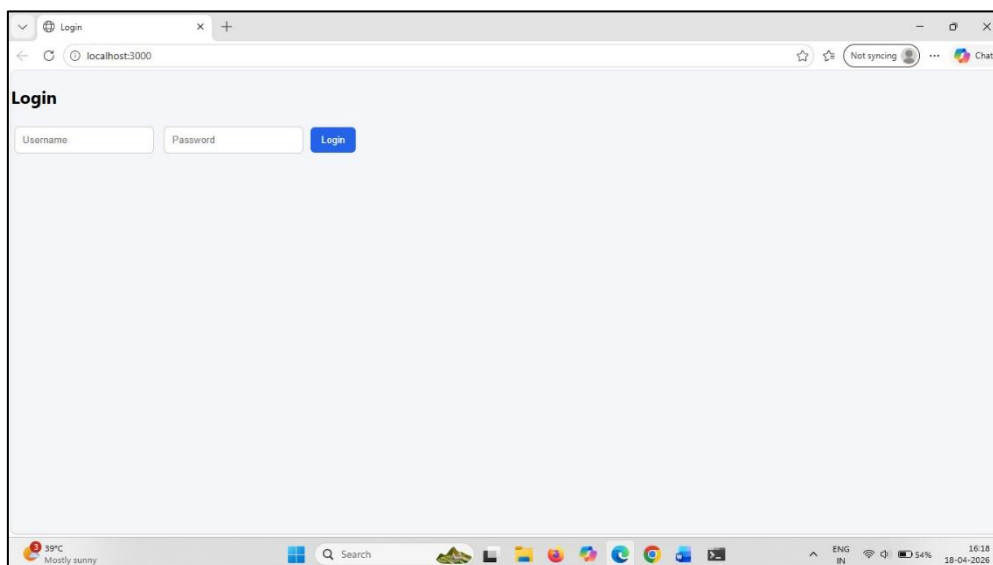


Figure 8.1: Login Module

8.2 Dashboard Module

The Dashboard Module provides an overview of the Vehicle Service Centre Management System. It acts as the main interface after login and allows users to navigate to different sections such as customer management, vehicle management, service details, and billing. The dashboard displays basic information and provides quick access to all major functionalities of the system.

Sample code:

Index.html

```
<h1>Dashboard</h1>

<nav>

  <a href="customer.html">Customers</a>

  <a href="vehicle.html">Vehicles</a>

  <a href="service.html">Services</a>

  <a href="billing.html">Billing</a>

  <a href="index.html">Logout</a>

</nav>
```

Sample Output:

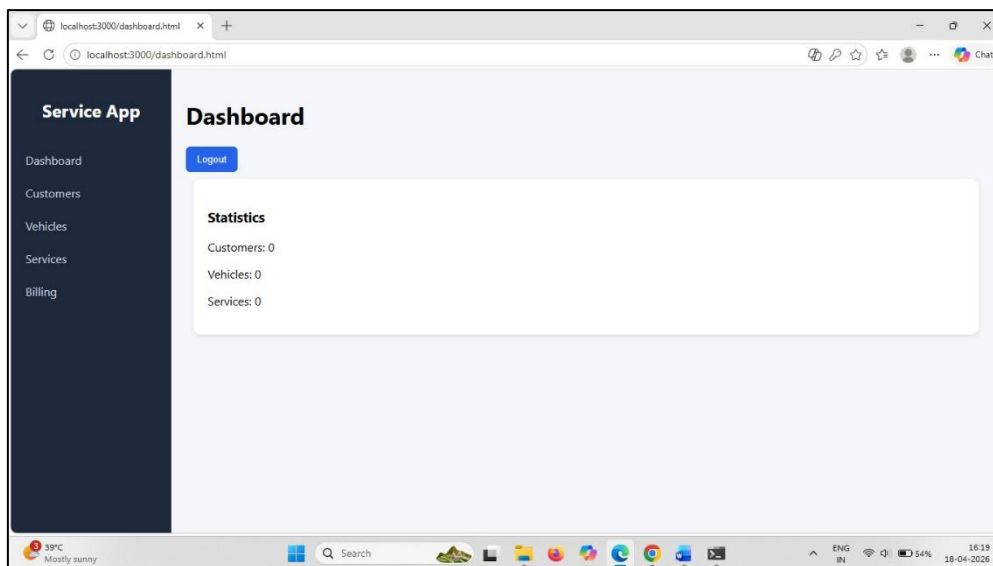


Figure 8.2: Dashboard Module

8.3 Customer Management Module

The Customer Management Module is used to manage customer details in the Vehicle Service Centre Management System. It allows users to add and view customer information such as name and contact number. This module helps in maintaining a proper record of all customers and improves service management.

Sample code:

Script.js

```
function addCustomer() {  
  
    let name = document.getElementById("name").value;  
  
    let phone = document.getElementById("phone").value;  
  
    let li = document.createElement("li");  
  
    li.innerText = name + " - " + phone;  
  
    document.getElementById("customerList").appendChild(li);  
  
}
```

Sample Output:

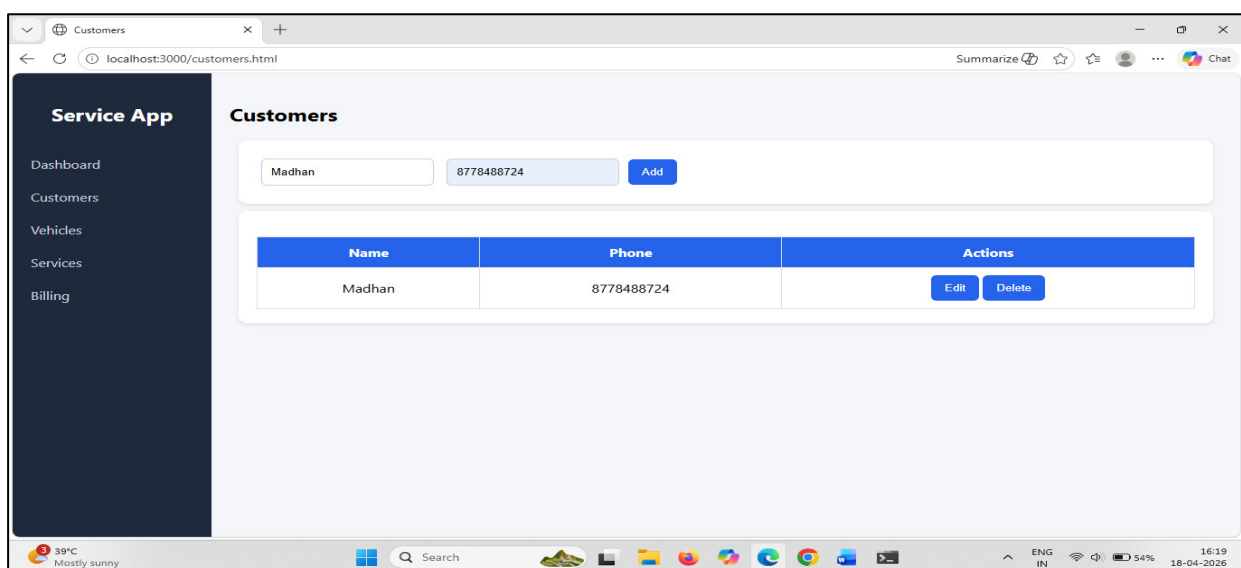


Figure 8.3: Customer Management Module

8.4 Vehicle Management Module

The Vehicle Management Module is used to manage vehicle details in the Vehicle Service Centre Management System. It allows users to add and view vehicle information such as vehicle model and registration number. This module helps in tracking vehicles that come for servicing and maintaining proper records.

Sample code:

Script.js

```
function addVehicle() {  
  
    let model = document.getElementById("model").value;  
  
    let number = document.getElementById("number").value;  
  
    let li = document.createElement("li");  
  
    li.innerText = model + " - " + number;  
  
    document.getElementById("vehicleList").appendChild(li);  
  
}
```

Sample Output:

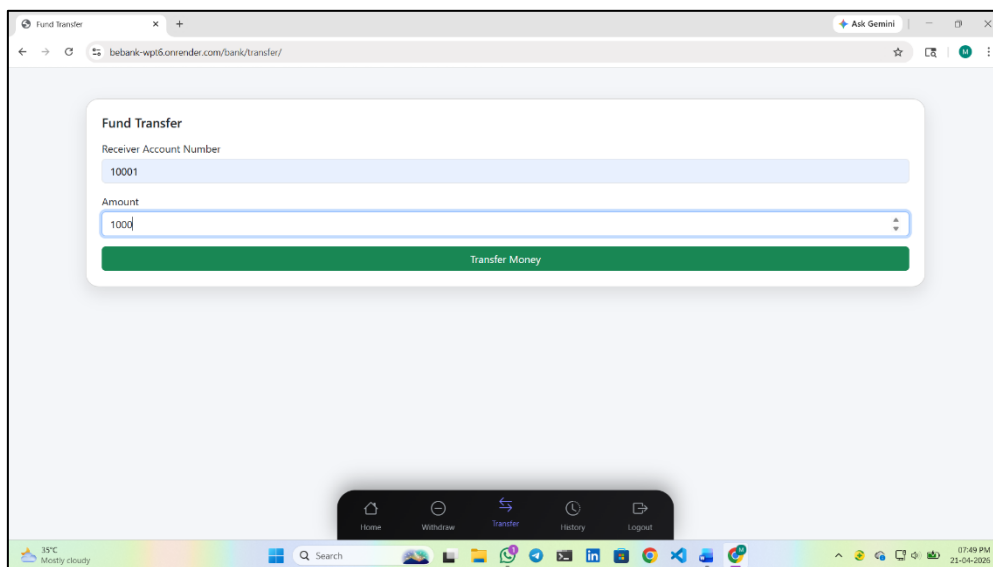


Figure 8.4: Vehicle Management Module

8.5 Service Management Module

The Service Management Module is used to manage service details in the Vehicle Service Centre Management System. It allows users to add and view different types of services such as oil change, washing, repair, and maintenance. This module helps in organizing service records and ensures efficient handling of service operations.

Sample code:

Script.js

```
function addService() {  
  
    let service = document.getElementById("service").value;  
  
    let cost = document.getElementById("cost").value;  
  
    let li = document.createElement("li");  
  
    li.innerText = service + " - ₹" + cost;  
  
    document.getElementById("serviceList").appendChild(li);  
  
}
```

Sample Output:

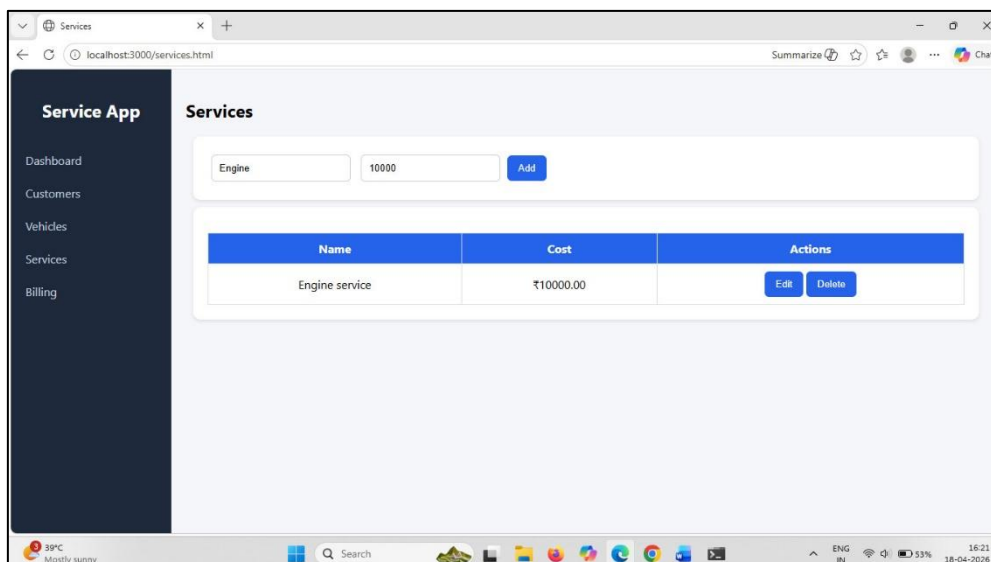


Figure 8.5: Service Management Module

8.6 Billing Module

The Billing Module is responsible for generating the total cost of services in the Vehicle Service Centre Management System. It calculates the bill based on the service charges entered and displays the final amount to the user. This module helps in maintaining accurate billing and improves transparency in payment processing.

Sample code:

Script.js

```
function generateBill() {  
    let cost = document.querySelectorAll("#serviceList li");  
    let total = 0;  
    cost.forEach(function(item) {  
        let price = item.innerText.split("₹")[1];  
        total += parseInt(price);  
    });  
    document.getElementById("totalAmount").innerText = "Total: ₹" + total;  
}
```

Sample Output:

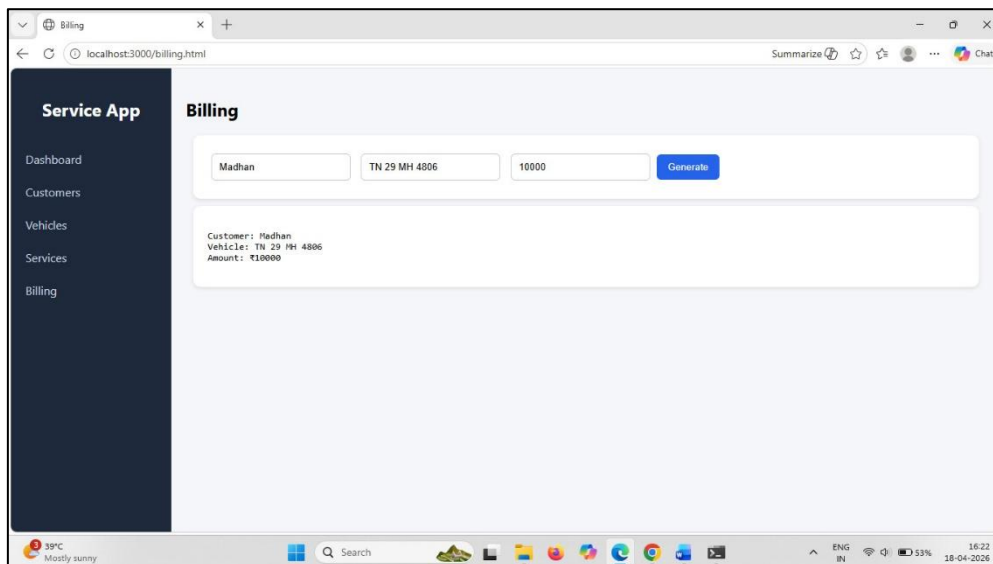


Figure 8.6: Billing Module

Chapter 9

TESTING

Testing is a crucial phase in software development used to verify whether the system functions correctly and meets the required objectives. It ensures that each module of the Vehicle Service Centre Management System performs as expected, handles data accurately, and produces reliable outputs. In this project, testing plays an important role because the system involves operations such as user login, customer management, vehicle handling, service processing, and billing generation. If these functionalities are not tested properly, errors in one module may affect the overall system performance and data accuracy.

Testing also helps in identifying bugs at an early stage and improves the quality, stability, and reliability of the application. It ensures that the system behaves correctly for both valid and invalid inputs and that all modules work together without failure. In this project, testing was conducted not only to verify individual components but also to validate the complete workflow—from user login to managing services and generating bills successfully. Therefore, the testing phase ensures that the system is accurate, reliable, and suitable for real-world service centre operations.

The testing phase consists of the following types:

1. Test Case Generation
2. Black Box Testing
3. White Box Testing
4. Unit Testing
5. Integration Testing
6. System Testing
7. Acceptance Testing

9.1 Test Case Generation

Test case generation is the process of designing test cases to verify whether each module of the Vehicle Service Centre Management System works according to the requirements. It ensures proper validation of inputs, processing logic, and output results. Test cases were created for major modules such as login, dashboard navigation, customer management, vehicle handling, service processing, and billing.

Sample Test Cases

Test Case ID	Module	Input	Expected Output	Actual Output	Result
TC01	Login	Valid username and password	Login successful	Login successful	Pass
TC02	Login	Invalid password	Error message displayed	Error message displayed	Pass
TC03	Customer	Valid customer details	Customer added successfully	Customer added successfully	Pass
TC04	Vehicle	Valid vehicle details	Vehicle added successfully	Vehicle added successfully	Pass
TC05	Service	Valid service name and cost	Valid service name and cost	Valid service name and cost	Pass
TC06	Billing	Multiple services selected	Total bill calculated correctly	Total calculated correctly	Pass
TC07	Dashboard	Click navigation links	Correct page opened	Correct page opened	Pass

Table 9.1: Sample Test Cases

9.2 Black Box Testing

Black box testing is a method of testing where the internal code structure is not considered, and the focus is entirely on validating the functionality of the system based on input and output behaviour. In the Vehicle Service Centre Management System, black box testing was performed on important modules such as login authentication, customer management, vehicle handling, service processing, and billing.

This testing ensured that the system provides correct outputs for valid inputs and displays appropriate error messages for invalid or incorrect inputs. It also verified that the user interface behaves as expected and that all user-facing features function correctly according to the project requirements. By applying black box testing, the system's external behaviour was validated effectively without analysing internal logic.

Sample Black Box Test Cases

Test Case ID	Input Condition	Expected Output	Actual Output	Result
BB01	Valid login credentials	Dashboard opens	Dashboard opens	Pass
BB02	Invalid password	Error message	Error message	Pass
BB03	Valid customer details	Customer added successfully	Customer added successfully	Pass
BB04	Invalid vehicle number	Error message displayed	Error message displayed	Pass
BB05	Valid service and cost	Service added successfully	Service added successfully	Pass
BB06	Multiple services selected	Total bill calculated correctly	Total calculated correctly	Pass

Table 9.2: Sample Black Box Test Cases

9.3 White Box Testing

White box testing is a technique that focuses on examining the internal logic, code structure, and control flow of the system. It is used to verify whether the implemented code executes correctly for different conditions and logical paths. In the Vehicle Service Centre Management System, white box testing was applied to critical components such as login validation, customer and vehicle data handling, service processing logic, and billing calculations.

This testing method helped in verifying conditional statements, loops, and data processing logic to ensure that the system produces accurate results under all scenarios. It also ensured that errors are handled properly and that there are no logical flaws in the implementation. By performing white box testing, the internal functionality of the system was validated thoroughly, improving overall code quality and reliability.

Sample White Box Test Cases

Test Case ID	Code Logic / Condition	Expected Output	Actual Output	Result
WB01	Validate login logic	Login allowed	Login allowed	Pass
WB02	Invalid password check	Error returned	Error returned	Pass
WB03	Customer data validation	Valid data accepted	Valid data accepted	Pass
WB04	Service cost calculation	Correct total calculated	Correct total calculated	Pass
WB05	Data display logic	Data displayed correctly	Data displayed correctly	Pass

Table 9.3: Sample White Box Test Cases

9.4 Unit Testing

Unit testing is performed to verify individual modules or components of the system independently. It ensures that each part of the system functions correctly without relying on other modules. In the Vehicle Service Centre Management System, unit testing was carried out on modules such as login validation, customer management, vehicle handling, service processing, and billing generation. Each module was tested separately to check whether it produces the expected output for given inputs.

This approach helps in identifying errors at an early stage and simplifies debugging, as issues can be traced back to specific modules. By ensuring that each component works correctly on its own, unit testing improves the stability and reliability of the overall system.

Sample Unit Test Cases

Test Case ID	Unit / Module	Expected Output	Actual Output	Result
UT01	Login validation	Login success	Login success	Pass
UT02	Customer Management Module	Customer added successfully	Customer added successfully	Pass
UT03	Vehicle module	Vehicle added successfully	Vehicle added successfully	Pass
UT04	Service module	Service added successful	Service added successful	Pass
UT05	Billing module	Total calculated correctly	Total calculated correctly	Pass

Table 9.4: Sample Unit Test Cases

9.5 Integration Testing

Integration testing is performed to verify whether multiple modules of the system work together correctly as a unified system. It focuses on checking the interaction between different components such as user interface, application logic, and data handling. In the Vehicle Service Centre Management System, integration testing was carried out between modules like login and dashboard, customer management and vehicle handling, service processing and billing, and overall data flow between modules.

This testing ensured that data flows correctly between modules and that there are no communication failures or inconsistencies. It also verified that the system performs operations smoothly when multiple components are combined. Integration testing plays an important role in ensuring that the system functions as a complete and connected application.

Sample Integration Test Cases

Test Case ID	Integrated Modules	Expected Output	Actual Output	Result
IT01	Login + Dashboard	Dashboard opens	Dashboard opened	Pass
IT02	Customer + Vehicle	Data linked correctly	Data linked correctly	Pass
IT03	Service + Billing	Total bill calculated	Total calculated correctly	Pass
IT04	All modules combined	System works smoothly	System works smoothly	Pass

Table 9.5: Sample Integration Test Cases

9.6 System Testing

System testing is conducted on the complete integrated system to evaluate whether the entire application works correctly as a whole. It validates the overall functionality, performance, and behaviour of the Vehicle Service Centre Management System after integrating all modules. In this project, system testing was performed to verify the complete workflow, including user login, dashboard navigation, customer and vehicle management, service processing, billing generation, and logout functionality.

This testing ensured that all modules operate together without errors and that the system meets the specified requirements. It also helped in identifying issues related to performance, data handling, and user interaction. By performing system testing, the application was validated in a real-world scenario to ensure smooth and reliable operation.

Sample System Test Cases

Test Case ID	System Operation	Expected Output	Actual Output	Result
ST01	Login process	Dashboard opens	Dashboard opened	Pass
ST02	Add customer	Customer added successfully	Customer added successfully	Pass
ST03	Add vehicle	Vehicle added successfully	Vehicle added successfully	Pass
ST04	Service process	Service added successfully	Service added successfully	Pass
ST05	Billing process	Total bill calculated	Total calculated correctly	Pass
ST06	Logout	Session ends	Session ended	Pass

Table 9.6: Sample System Test Cases

9.7 Acceptance Testing

Acceptance testing is the final stage of testing conducted to determine whether the system meets user requirements and is ready for deployment. It focuses on validating the system from the user's perspective and ensuring that it fulfils all functional and operational needs. In the Vehicle Service Centre Management System, acceptance testing was carried out to verify features such as secure login, dashboard access, customer and vehicle management, service handling, and billing operations.

This testing ensured that the system behaves as expected and provides a user-friendly experience. It also confirmed that the application achieves its objective of providing an efficient and reliable service management platform. Acceptance testing ultimately ensures that the system is suitable for real-world use and meets the expectations of end users.

Sample Acceptance Test Cases

Test Case ID	User Requirements	Expected Output	Actual Output	Result
AT01	Secure login	Login success	Login successful	Pass
AT02	Customer management	Customer handled correctly	Customer handled correctly	Pass
AT03	Vehicle management	Vehicle handled correctly	Vehicle handled correctly	Pass
AT04	Service and billing	Bill generated correctly	Bill generated correctly	Pass

Table 9.7: Sample Acceptance Test Cases

Chapter 10

CONCLUSION

The Vehicle Service Centre Management System was developed to provide a centralized, efficient, and user-friendly solution for managing service centre operations through a web-based platform. The system supports key functionalities such as user login, dashboard navigation, customer management, vehicle tracking, service handling, and billing generation. By integrating all these operations into a single system, it ensures that service activities are performed in a structured and organized manner, reducing complexity and improving usability.

The project significantly reduces manual work and enhances the accuracy of service management by automating essential processes. It ensures that customer details, vehicle records, service information, and billing data are stored and managed reliably, minimizing the chances of human errors. The login system provides controlled access to the system, ensuring that only authorized users can perform operations.

Another important outcome of the project is the improvement in user experience and accessibility. Users can easily navigate through the system, manage customer and vehicle details, add services, and generate bills using a simple interface. This makes the system easy to use while still being effective for handling real-world service centre tasks.

Overall, the Vehicle Service Centre Management System successfully achieves its objective of creating a reliable and efficient platform for managing service centre operations. It improves operational efficiency, simplifies data management, and ensures better service delivery. The system can be further enhanced with advanced features, making it a scalable solution for future development.

In addition, the system provides a strong foundation for future enhancements such as online booking, automated service reminders, payment integration, and real-time reporting. These improvements can further increase efficiency and customer satisfaction, making the system more suitable for large-scale service centres and modern business environments.

Chapter 11

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